

## History of jQuery

jQuery is a **fast, small, and feature-rich JavaScript library** that simplifies HTML document traversal, event handling, animation, and Ajax interactions.

## Who Created jQuery?

- **Creator:** John Resig
- **Introduced:** January 2006
- **First Released:** jQuery 1.0 on **August 26, 2006**
- **Website:** [jquery.com](http://jquery.com)

## Why jQuery Was Popular

- Cross-browser compatibility
- Easy DOM manipulation
- Simple syntax (`$("#id")`)
- AJAX and animations made simple
- Large plugin ecosystem



## Current Status (as of 2025)

- jQuery is still maintained and used in legacy projects
- Latest version: **jQuery 3.x** (still actively patched for security and bugs)
- Modern apps mostly use native JavaScript or frameworks



## Basic jQuery Syntax

```
$(selector).function();  
jQuery(selector).function();
```

**\$** or **jQuery** — Access jQuery functionalities

**selector** — Selects HTML elements

**function()** — Performs an action on the element(s)



**\$(document).ready()** in jQuery

♦ **What is \$(document).ready()?**

`$(document).ready()` is a jQuery method that runs your JavaScript **only after the HTML document has been fully loaded and parsed** — **before images, CSS, and other external content are loaded.**

```
$(document).ready(function() {  
    // jQuery code goes here  
});
```

OR (shorthand):

```
$(function() {  
    // jQuery code goes here  
});
```

## Why Use It?

- Ensures the DOM is fully loaded before your code runs
- Prevents errors like "element not found"
- Especially useful when scripts are placed in the `<head>`

## **Jquery Selectors:**

jQuery selectors allow you to select and manipulate HTML elements easily using CSS-like syntax. They are one of the core features of jQuery.

## **Types of jQuery Selectors**

### **1. Basic Selectors**

| Selector                                                                   | Example                  | Description                              |
|----------------------------------------------------------------------------|--------------------------|------------------------------------------|
| <code>*</code>                                                             | <code>\$("*")</code>     | Selects all elements                     |
| <code>#id</code>                                                           | <code>\$("#box")</code>  | Selects element with specific ID         |
| <code>.class</code>                                                        | <code>\$(".item")</code> | Selects all elements with specific class |
| <code>Element</code>                                                       | <code>\$("p")</code>     | Selects all <p> tags                     |
| Multiple element, <code>\$("h1, h2")</code> Selects multiple element types |                          |                                          |
| ancestor descendant                                                        | <code>\$("div p")</code> | Selects all nested descendants           |

## **Jquery events:**

jQuery provides powerful methods to handle events like clicks, key presses, mouse movements, form submissions, and more — with simplified syntax compared to plain JavaScript.

## **Common jQuery Events**

| Event Method                                            | Description                        |
|---------------------------------------------------------|------------------------------------|
| <code>.click()</code>                                   | User clicks an element             |
| <code>.dblclick()</code>                                | User double-clicks an element      |
| <code>.hover()</code>                                   | Mouse enters and leaves an element |
| <code>.keydown()</code> / <code>.keyup()</code>         | Key pressed or released            |
| <code>.submit()</code>                                  | Form is submitted                  |
| <code>.change()</code>                                  | Value of input/select changes      |
| <code>.focus()</code> / <code>.blur()</code>            | Element gains or loses focus       |
| <code>.mouseenter()</code> / <code>.mouseleave()</code> | Mouse enters or leaves an element  |

## **Examples of jQuery Events**

### 1. Click Event

```
<button id="myBtn">Click Me</button>
<p id="msg"></p>

<script>
  $(document).ready(function() {
    $("#myBtn").click(function() {
```

```
        $("#msg").text("Button was clicked!");  
    });  
});  
</script>
```

## 2. Mouse Events : hover()

```
<div id="hoverBox" style="width:150px; height:100px;  
background:lightblue;"></div>  
  
<script>  
    $(document).ready(function() {  
        $("#hoverBox").hover(  
            function() { $(this).css("background", "yellow"); }, // mouse enter  
            function() { $(this).css("background", "lightblue"); } // mouse  
leave  
        );  
    });  
</script>
```

## 3. Keyboard Events: keydown()

```
<input type="text" id="inputField">  
<p id="keyLog"></p>  
  
<script>  
    $(document).ready(function() {
```

```
$("#inputField").keydown(function(event) {  
    $("#keyLog").text("You pressed: " + event.key);  
});  
});  
</script>
```

#### 4. Form Submit Event

```
<form id="myForm">  
    <input type="text" name="username">  
    <button type="submit">Submit</button>  
</form>  
  
<p id="formMsg"></p>  
  
<script>  
$(document).ready(function() {  
    $("#myForm").submit(function(event) {  
        event.preventDefault(); // prevent page refresh  
        $("#formMsg").text("Form submitted!");  
    });  
});  
</script>
```

## 5. Change Event (Dropdown or Input)

```
<select id="colorSelect">
  <option value="Red">Red</option>
  <option value="Green">Green</option>
</select>

<p id="selectedColor"></p>

<script>
  $(document).ready(function() {
    $("#colorSelect").change(function() {
      let selected = $(this).val();
      $("#selectedColor").text("You selected: " + selected);
    });
  });
</script>
```

## 6. Focus and Blur

```
<input type="text" id="focusInput">

<script>
  $(document).ready(function() {
    $("#focusInput").focus(function() {
      $(this).css("background-color", "lightyellow");
    }).blur(function() {
      $(this).css("background-color", "");
    });
  });
</script>
```

```
});  
});  
</script>
```

## Bonus: Using .on() for Multiple Events

```
$("#myBtn").on("click mouseenter", function() {  
    alert("Mouse entered or button clicked!");  
});
```

## Jquery animation:

jQuery makes it easy to create animations and visual effects with just a few lines of code. You can show, hide, fade, slide, and even build custom animations. The `animate()` method in jQuery lets you **create custom animations** by gradually changing CSS properties with smooth transitions.

## Basic Example

```
<div id="box" style="width:100px; height:100px; background:blue;  
position:relative;"></div>  
<button id="start">Animate</button>  
  
<script>
```

```
$(document).ready(function() {  
    $("#start").click(function() {  
        $("#box").animate({  
            left: '250px',  
            height: '150px',  
            width: '150px',  
            opacity: 0.5  
        }, 1000);  
    });  
});  
</script>
```

## Properties You Can Animate

### Numeric properties only:

- left, top, height, width, margin, padding
- opacity
- fontSize (use em, px, %)

 You cannot animate: background-color, display, or non-numeric properties directly with `animate()`

### Example Use Cases

- **Animated banners**
- **Progress bars**
- **Sliding panels**
- **Custom loaders**

## **Jquery Effects:**

jQuery provides built-in effects and animations to enhance the visual interaction of your web pages — no need for complex JavaScript or CSS transitions!

## **Common jQuery Effects**

| Effect           | Method                                                                                                 | Description                      |
|------------------|--------------------------------------------------------------------------------------------------------|----------------------------------|
| Show / Hide      | <code>.show()</code> , <code>.hide()</code> , <code>.toggle()</code>                                   | Show or hide elements            |
| Fade in/out      | <code>.fadeIn()</code> , <code>.fadeOut()</code> , <code>.fadeToggle()</code> , <code>.fadeTo()</code> | Fade elements in/out             |
| Slide vertically | <code>.slideDown()</code> , <code>.slideUp()</code> , <code>.slideToggle()</code>                      | Slide elements vertically        |
| Custom           | <code>.animate()</code>                                                                                | Animate any numeric CSS property |

## 1. Show / Hide / Toggle

```
<p id="text">Hello jQuery</p>
<button id="hideBtn">Hide</button>
<button id="showBtn">Show</button>
<button id="toggleBtn">Toggle</button>

<script>
  $(document).ready(function() {
    $("#hideBtn").click(() => $("#text").hide());
    $("#showBtn").click(() => $("#text").show());
    $("#toggleBtn").click(() => $("#text").toggle());
  });
</script>
```

## 2. Fade Effects

```
<div id="box" style="width:100px; height:100px; background:red;"></div>
<button id="fadeBtn">Fade Out</button>

<script>
  $(document).ready(function() {
    $("#fadeBtn").click(() => $("#box").fadeOut(1000));
  });
</script>
```

## 3. Slide Effects

```
<div id="panel" style="display:none; background:lightblue;">Sliding
Panel</div>
<button id="slideBtn">Slide Toggle</button>

<script>
  $(document).ready(function() {
    $("#slideBtn").click(() => $("#panel").slideToggle(500));
  });
</script>
```

 Bonus: Chaining Effects

```
$("#box").slideUp(500).fadeIn(500).animate({ left: '100px' }, 400);
```

## Duration Values

- "slow" = 600ms
- "fast" = 200ms
- Or use milliseconds: 1000 = 1 second

## Jquery DOM traversal and Manipulation:

jQuery makes it very easy to navigate (traverse) the DOM and manipulate HTML elements, content, and attributes.

## ▲ Upward Traversal

| Method                               | Description                                                |
|--------------------------------------|------------------------------------------------------------|
| <code>.parent()</code>               | Selects the direct parent                                  |
| <code>.parents()</code>              | Selects all ancestors                                      |
| <code>.closest()</code>              | Selects the closest matching ancestor                      |
| <code>.parentsUntil(selector)</code> | Selects all ancestors up to but not including the selector |

Example:

```
$("#child").parent().css("border", "1px solid red");
```

## ▼ Downward Traversal

| Method                   | Description                       |
|--------------------------|-----------------------------------|
| <code>.children()</code> | Selects direct children           |
| <code>.find()</code>     | Selects descendants (deep search) |

Example:

```
$("#list").children("li").css("color", "blue");  
$("#container").find("p").css("font-weight", "bold");
```

## Sideways Traversal

| Method                   | Description                  |
|--------------------------|------------------------------|
| <code>.siblings()</code> | Selects all siblings         |
| <code>.next()</code>     | Selects the next sibling     |
| <code>.prev()</code>     | Selects the previous sibling |
| <code>.nextAll()</code>  | All next siblings            |
| <code>.prevAll()</code>  | All previous siblings        |

### Example:

```
$("#item1").next().css("color", "green");  
$("#item1").siblings().css("color", "gray");
```

## 2. DOM Manipulation in jQuery

| Method               | Description             |
|----------------------|-------------------------|
| <code>.html()</code> | Get or set HTML content |
| <code>.text()</code> | Get or set text content |

```
$("#info").html("<b>Bold Text</b>");  
$("#info").text("Just plain text");
```



## 1. .html()

Gets or sets the **HTML content**.

```
$("#myDiv").html("<b>Hello World!</b>");
```

## 2. .text()

Gets or sets the **text content**.

```
$("#myDiv").text("Just plain text");
```

## 3. .val()

Gets or sets the **value** of form elements.

```
$("#input").val("New Value");
```

## 4. .attr()

Gets or sets **attributes**.

```
$("#img").attr("src", "new-image.jpg");
```

## 5. .css()

Gets or sets **CSS styles**.

```
$("#p").css("color", "red");
```

## Adding/Removing Elements

### 1. .append() / .prepend()

Adds content **inside** an element.

```
$("#myDiv").append("<p>Appended</p>");  
$("#myDiv").prepend("<p>Prepended</p>");
```

### 2. .after() / .before()

Adds content outside an element.

```
$("#myDiv").after("<hr>");  
$("#myDiv").before("<h3>Heading</h3>");
```

### 3. `.remove()` / `.empty()`

- `remove()` deletes the element.
- `empty()` clears its contents.

```
$("#myDiv").remove();  
$("#myDiv").empty();
```

#### ✓ Example Use Case

```
<ul>  
  <li id="item1">Apple</li>  
  <li>Banana</li>  
  <li>Cherry</li>  
</ul>  
  
<script>  
  $("#item1").next().css("color", "red"); // Banana becomes red  
  $("li").last().text("Mango");           // Cherry changes to Mango  
</script>
```

## Data Attributes and Templating :

## ✓ What are Data Attributes?

Custom HTML attributes starting with **data-**, used to store **extra information**.

```
<div id="user" data-id="101" data-role="admin">John Doe</div>
```

## 🔍 Accessing Data Attributes

➤ Using **.data()** in jQuery:

```
let userId = $("#user").data("id"); // returns 101
let userRole = $("#user").data("role"); // returns "admin"
```

## 🔧 Setting Data Attributes

```
$("#user").data("role", "editor"); // Updates the data-role
```

But this won't reflect in the HTML. To update the actual HTML:

```
$("#user").attr("data-role", "editor");
```

## 📋 Use Case Example:

```
<button class="btn" data-user-id="5">View</button>

<script>
  $(".btn").click(function() {
    let id = $(this).data("user-id");
    alert("User ID: " + id);
  });
</script>
```

## 2. Templates in jQuery

### What is Templating?

Templating means **inserting data into HTML structure dynamically**.

jQuery doesn't have built-in templating, but you can use basic approaches like:

- ♦ Method 1: Using String Templates

```
<div id="output"></div>

<script>
  let user = { name: "Alice", age: 25 };

  let template = `<p>Name: ${user.name}</p><p>Age: ${user.age}</p>`;
  $("#output").html(template);
</script>
```

- ♦ Method 2: Using `.clone()` from Hidden Template

```
<div id="template" style="display: none;">
  <div class="user">
    <h4 class="name"></h4>
    <p class="age"></p>
  </div>
</div>

<div id="user-list"></div>

<script>
  let users = [
    { name: "John", age: 30 },
    { name: "Sara", age: 22 }
  ];

  users.forEach(function(user) {
    let item = $("#template .user").clone();
    item.find(".name").text(user.name);
    item.find(".age").text("Age: " + user.age);
    $("#user-list").append(item);
  });
</script>
```

## jQuery DOM Utility Functions

These utility methods are commonly used for checking, modifying, or working with DOM elements.

## ♦ 1. `$.isEmptyObject(obj)`

✓ Checks if an object has **no properties**.

```
let obj = {};  
$.isEmptyObject(obj); // true
```

## ♦ 2. `$.isArray(obj)`

✓ Checks if the passed object is an array.

```
let arr = [1, 2, 3];  
$.isArray(arr); // true
```

## ♦ 3. `$.trim(str)`

✓ Removes **leading and trailing whitespace**.

```
let text = "  hello world  ";
```

```
let trimmed = $.trim(text); // "hello world"
```

## ♦ 4. \$.type(obj)

✓ Returns a **string representing the type** of the object (more accurate than `typeof`).

```
$.type(123);           // "number"  
$.type("hello");       // "string"  
$.type([1, 2, 3]);     // "array"  
$.type({});            // "object"
```

## ♦ 5. \$.isFunction(obj)

✓ Checks if the argument is a **function**.

```
function greet() {}  
$.isFunction(greet); // true
```

## ♦ 6. \$.each(obj, callback)

✓ Iterates over arrays or objects.

```
$.each([10, 20, 30], function(index, value) {  
    console.log(index + ": " + value);  
});
```

## What Are jQuery Plugins?

jQuery plugins are reusable chunks of code that extend the functionality of jQuery. They let you add features like sliders, popups, form validation, and animations without rewriting code.

## **Using jQuery Plugins**

✓ Steps to Use a Plugin:

1. Include jQuery
2. Include the Plugin's JS and CSS files
3. Initialize the plugin

 Example: Using a Slider Plugin (e.g., Slick Slider)

```
<!-- Include jQuery -->
```

```
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>

<!-- Include Plugin CSS and JS -->
<link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/slick-carousel@1.8.1/slick/slick.css"/>
<script
src="https://cdn.jsdelivr.net/npm/slick-carousel@1.8.1/slick/slick.min.js"
></script>

<!-- HTML -->
<div class="slider">
  <div>Slide 1</div>
  <div>Slide 2</div>
  <div>Slide 3</div>
</div>

<!-- Initialization -->
<script>
$(document).ready(function(){
  $('.slider').slick({
    autoplay: true,
    dots: true
  });
});
</script>
```

## Popular jQuery Plugins

[www.php-training.in](http://www.php-training.in) / [www.phpcourse.in](http://www.phpcourse.in)

**Mulund :** 303,304,309 Shree Samarth Plaza , RRT Road, Opp. Mulund Station . Mulund West . Mumbai 400080

**Vile Parle :** Datapoint computers: vijaya Apt , Ground Floor ,Bajaj Road , Vile parle West, Vile Parle (West) . MUMBAI -56



| Plugin         | Name | Purpose                                 |
|----------------|------|-----------------------------------------|
| Slick          |      | Responsive <b>image</b> /content slider |
| jQuery UI      |      | Widgets like datepicker, <b>dialog</b>  |
| Lightbox2      |      | <b>Image</b> lightbox gallery viewer    |
| DataTables     |      | Interactive table sorting/search        |
| Validate.js    |      | Form validation                         |
| Magnific Popup |      | Lightweight popup modals                |
| Waypoints      |      | Trigger functions on scroll             |

## Creating Your Own jQuery Plugin

**You can define a custom plugin using:**

```
(function($) {  
    $.fn.highlight = function(color) {  
        this.css("background-color", color || "yellow");  
        return this;  
    };  
})(jQuery);
```

## How to Use Your Plugin:

```
<p>Hello World!</p>  
  
<script>  
    $("p").highlight("lightblue");  
</script>
```

**Q: Write a jQuery program to get a single element from a selection of elements of a HTML page.**

Here is a simple jQuery program to get a single element from a selection of elements on an HTML page — using `.eq(index)`:

## Goal:

Select a specific element (e.g., the **2nd paragraph**) from multiple `<p>` tags and manipulate it.

## HTML Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>jQuery Select Single Element</title>
  <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
</head>
<body>

  <p>Paragraph 1</p>
  <p>Paragraph 2</p>
  <p>Paragraph 3</p>

  <script>
    // Wait for the DOM to be ready
    $(document).ready(function(){
      // Select the 2nd paragraph (index 1) and change its text color
```

```
        $("p").eq(1).css("color", "red").text("This is the selected  
paragraph.");  
    });  
</script>  
  
</body>  
</html>
```

## Explanation:

- `$("p")` selects all `<p>` tags.
- `.eq(1)` gets the **second** element (index starts at 0).
- `.css()` and `.text()` apply styling and modify the text.

**Q: You are having sample data for the link. Write jQuery code to change the hyperlink and the text of an existing link.**

Certainly! Here's how you can use jQuery to change the hyperlink (`href`) and the text of an existing link (`<a>` tag) using sample data.

## HTML Example:

```
<!DOCTYPE html>  
<html>  
<head>
```

```
<title>Change Link with jQuery</title>
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
</head>
<body>

<a id="myLink" href="https://oldlink.com">Old Link</a>

<script>
  $(document).ready(function(){
    // Sample new data
    let newURL = "https://www.example.com";
    let newText = "Visit Example";

    // Change the href and the text
    $("#myLink").attr("href", newURL).text(newText);
  });
</script>

</body>
</html>
```

## Explanation:

- `$("#myLink")` selects the `<a>` tag with id `myLink`.
- `.attr("href", newURL)` updates the link destination.
- `.text(newText)` changes the visible text.

**Q: Write a jQuery program to attach a click and double-click events to all <p> elements.**

Here's a jQuery program to attach both **click** and **double-click** events to all **<p>** elements on an HTML page.

 **Goal:**

- On **click**: Change the text color to **blue**
- On **double-click**: Change the background color to **yellow**

 **Full HTML + jQuery Code:**

```
<!DOCTYPE html>
<html>
<head>
  <title>Click and Double Click Events</title>
  <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
  <style>
    p {
      cursor: pointer;
      padding: 10px;
    }
  </style>
</head>
<body>

  <p>Paragraph 1</p>
```

```
<p>Paragraph 2</p>
<p>Paragraph 3</p>

<script>
    $(document).ready(function(){
        // Click event: change text color
        $("p").click(function(){
            $(this).css("color", "blue");
        });

        // Double-click event: change background
        $("p").dblclick(function(){
            $(this).css("background-color", "yellow");
        });
    });
</script>

</body>
</html>
```

## Explanation:

- `$("p").click(...)`: Attaches click event to all `<p>` elements.
- `$("p").dblclick(...)`: Attaches double-click event to the same elements.
- `$(this)`: Refers to the clicked paragraph.



[Mulund , Vile Parle]

o Also find the position of the mouse pointer relative to the left and top edges of the document.

1. Hides all headings (`<h1>`, `<h2>`, etc.) when they are clicked.
2. Displays the mouse pointer's position (relative to the top-left corner of the document).

```
<!DOCTYPE html>
<html>
<head>
  <title>Hide Headings & Show Mouse Position</title>
  <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
  <style>
    #mousePosition {
      position: fixed;
      top: 10px;
      left: 10px;
      background-color: #f0f0f0;
      padding: 5px;
      border: 1px solid #ccc;
      font-family: Arial;
    }
  </style>

```

```
</style>
</head>
<body>

<h1>Heading One</h1>
<h2>Heading Two</h2>
<h3>Heading Three</h3>
<div id="mousePosition">Mouse Position: X: 0, Y: 0</div>

<script>
$(document).ready(function() {
    // Hide any heading on click
    $("h1, h2, h3, h4, h5, h6").click(function() {
        $(this).hide();
    });

    // Show mouse position relative to the document
    $(document).mousemove(function(event) {
        let x = event.pageX;
        let y = event.pageY;
        $("#mousePosition").text("Mouse Position: X: " + x + ", Y: " + y);
    });
});
</script>

</body>
</html>
```

## Explanation:

- `$("h1, h2, h3, h4, h5, h6").click(...)`: Binds a click event to all heading tags.
- `$(this).hide()`: Hides the clicked heading.
- `$(document).mousemove(...)`: Tracks mouse movements.
- `event.pageX / event.pageY`: Gives the mouse position **relative to the document**, not the viewport.